

Chapter 5: Structured Knowledge Representation

4 hours | 7 marks

"Knowledge has a structure. If we represent it well, reasoning becomes trivial."

— Marvin Minsky

The Hook — How Do You Store "What You Know"?

Logic (Chapter 4) is powerful but verbose. Writing everything as FOPL sentences is tedious.

Consider trying to represent everything you know about a "bird":

- It has wings
- It has a beak
- It can fly (usually)
- It lays eggs
- It has feathers
- It is an animal
- It has a Latin name
- It has a mass
- It has a habitat...

Logic would require a separate sentence for each fact. What if we could organize knowledge more naturally — like a **mental model** or a **filing cabinet**?

That's the goal of **structured knowledge representation**.

5.1 Representations and Mappings

The Core Challenge

"How do we translate the real world into a form a machine can reason about?"

This mapping has two components:

REAL WORLD	REPRESENTATION
Objects	→ Symbols/Nodes
Properties	→ Attributes/Slots
Relations	→ Links/Arcs
Events	→ State changes

Any representation must balance:

Concern	Trade-off
Expressiveness	More powerful = harder to compute
Efficiency	Faster = less expressive
Completeness	Can we represent everything needed?
Naturalness	Does it match human intuition?

5.2 Approaches to Knowledge Representation

There are four main paradigms in AI:

Approach 1: Logical (Declarative)

What we covered in Chapter 4 — facts and rules in formal logic.

```
All(x: Bird(x) → CanFly(x))  
Bird(Tweety)  
→ CanFly(Tweety)
```

Pros: Rigorous, supports automated reasoning

Cons: Verbose, doesn't handle exceptions well

Approach 2: Procedural

Knowledge stored as **how to do something** — procedures and algorithms.

```
HOW_TO: Find_Shortest_Path(graph, start, goal):  
    1. Initialize open_list with start  
    2. While open_list not empty...  
    ...
```

Pros: Efficient for known procedures

Cons: Hard to generalize or modify

Approach 3: Network-Based

Knowledge stored as a **graph** — nodes connected by labeled relationships.

This is where **Semantic Nets** live.

Approach 4: Structured Objects

Knowledge stored as **structured records** with named slots.

This is where **Frames** live.

5.3 Issues in Knowledge Representation

Before diving into specific methods, understand the challenges:

Issue 1: The Frame Problem

"When I pick up a block, what else changes?"

In the real world, we implicitly know: moving a block doesn't change the color of the sky, the temperature of the room, etc. But a machine needs to be told explicitly what stays the same.

The frame problem: How to represent which things stay unchanged when an action is performed?

Issue 2: The Qualification Problem

Rules have exceptions:

- "Birds can fly" — except penguins, ostriches, dead birds, birds with broken wings...

How many qualifications do we add before we've covered everything?

Issue 3: Closed World Assumption (CWA)

If we don't know something is true, is it false?

CWA: If the knowledge base doesn't contain P , assume $\neg P$.

Example: "Who flies to Kathmandu on Sundays?" If the schedule doesn't list it $\rightarrow$ no one does.

This works for databases but fails for open-world reasoning.

Issue 4: The Completeness Problem

Can we ever represent ALL knowledge? No — knowledge representation is always a useful **partial model**, not a complete one.

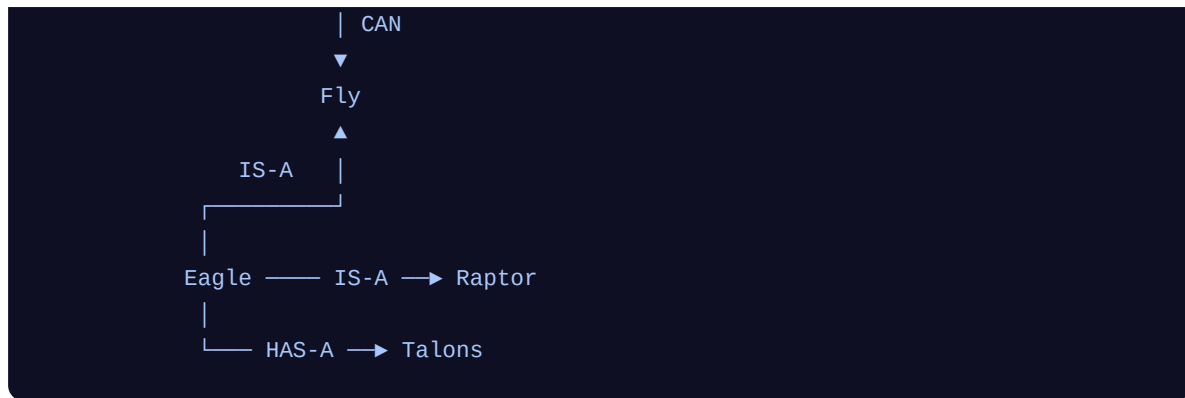
5.4 Semantic Networks

What Are They?

A **semantic network** is a graph where:

- **Nodes** = concepts or objects
- **Arcs (edges)** = labeled relationships between concepts





Key Relationships in Semantic Networks

Relationship	Meaning	Example
IS-A	Class membership	Eagle IS-A Bird
AKO (A-Kind-Of)	Subclass	Bird AKO Animal
HAS-PART	Part-of relationship	Bird HAS-PART Wings
CAN	Capability	Bird CAN Fly
LOCATION	Where something is	Eagle LOCATION Mountain

Inheritance in Semantic Networks

The power of IS-A hierarchies: **properties are inherited down the hierarchy.**

```
Animal — HAS —> Heart
  ↑ IS-A
Bird — CAN —> Fly
  ↑ IS-A
Eagle
```

Eagle inherits from Bird:

- Eagle CAN Fly (inherited from Bird)
- Eagle HAS Heart (inherited from Animal via Bird)

We don't need to repeat "has heart" for every animal — inheritance handles it automatically!

Exception Handling in Semantic Networks

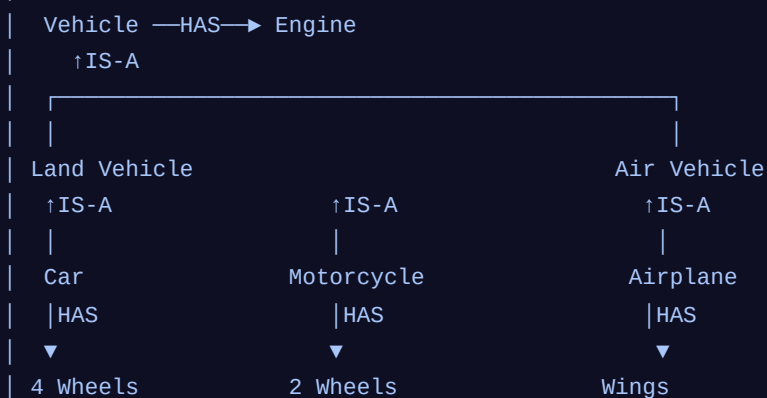
```

Animal — CAN → Breathe
  ↑ IS-A
Bird — CAN → Fly
  ↑ IS-A
Penguin — CANNOT → Fly (overrides inherited CAN Fly!)

```

The **most specific** rule wins — Penguin's local property overrides the inherited one.

A Complete Example: Transportation Network



From this network, we can infer:

- Airplane IS-A Vehicle → Airplane HAS-A Engine (inherited)
- Car IS-A Land Vehicle IS-A Vehicle → Car HAS-A Engine

Limitations of Semantic Networks

- Hard to represent **quantifiers** ("some", "all", "most")
- Hard to represent **negative** information
- Hard to represent **procedural** knowledge
- No standard notation — different systems use different labels
- Can become unclear for complex relationships

5.5 Frames

What Are They?

A **frame** is a structured record (like a class in programming) that represents a **stereotyped situation**.

Think of it as a template: "What does a typical X look like?"

FRAME: Bird	
SLOT	VALUE
IS-A	Animal
Body-Covering	Feathers
Locomotion	Flying, Walking
Reproduction	Egg-laying
Number-of-legs	2
Has-wings	True

Frame Hierarchy (Inheritance)

FRAME: Animal	
IS-A: Living-Thing	
Breathes: True	
Reproduction: varies	
↑ IS-A	
FRAME: Bird	
IS-A: Animal	
Body-Covering: Feathers	← overrides Animal's default
Locomotion: Flying, Walking	
Number-of-legs: 2	
↑ IS-A	
FRAME: Eagle	
IS-A: Bird	
Diet: Carnivore	← adds new slot
Habitat: Mountains	
Wingspan: 2 meters	

Eagle inherits from Bird and Animal. Its own slots **override** inherited ones where specified.

Slots and Facets

Each slot in a frame can have multiple **facets**:

```
FRAME: Person
  Slot: Age
    VALUE: (blank – to be filled)
    DEFAULT: 30
    CONSTRAINT: Integer, 0-150
    IF-ADDED: Update-age-dependent-calculations
    IF-REMOVED: Warn-about-data-deletion
    IF-NEEDED: Ask-user-for-age
```

Demons (IF-ADDED, IF-NEEDED, IF-REMOVED) make frames **active** — they can trigger procedures when slots change!

Frame Example: Medical Knowledge

```
FRAME: Patient
  Name: (value)
  Age: (value, default 0)
  Symptoms: (value, multiple allowed)
  Diagnosis: (inferred from symptoms)
  Treatment: (IF-NEEDED: call treatment-selection-procedure)

FRAME: COVID-19-Patient
  IS-A: Patient
  Symptoms: Fever, Cough, Fatigue
  Quarantine-required: True
  Treatment: Antiviral + Rest
```

When we create an instance of COVID-19-Patient with a name and age, it automatically inherits treatment protocols!

5.6 Conceptual Dependencies

The Idea

Developed by **Roger Schank** in the 1970s, Conceptual Dependency (CD) theory says:

"Any meaning that can be expressed in any human language can be represented using a small set of primitive actions."

Schank identified **~12 primitive actions** (ACTs):

Primitive	Meaning	Example
ATRANS	Transfer of abstract relationship	Give, take
PTRANS	Physical transfer of location	Go, carry
MTRANS	Transfer of mental information	Tell, read
MBUILD	Mental construction	Infer, decide
INGEST	Take into body	Eat, drink, breathe
EXPEL	Expel from body	Bleed, cry
PROPEL	Apply force	Hit, throw, push
MOVE	Move body part	Nod, wave hand
GRASP	Physical grip	Grab, release
SPEAK	Produce sounds	Say, sing

Example: "John gave Mary a book"

In CD notation:

```
John — ATRANS → book
      — FROM → John
      — TO → Mary
      — TIME → past
```

And "Mary gave John money":

```
Mary — ATRANS → money
      — FROM → Mary
      — TO → John
      — TIME → past
```

Notice: Although the English sentences look different, similar deep structures emerge for the same type of action.

Why This Matters

CD theory allows AI to:

- Understand that "John transferred the book to Mary" and "John gave Mary the book" mean the same thing
 - Make inferences ("If X AINGESTed food, then X is less hungry")
 - Translate between languages by working at the conceptual level
-

5.7 Scripts

What Are Scripts?

A **script** is a frame-like structure that represents a **stereotyped sequence of events** — a story template.

Developed by Schank and Abelson in their famous book "Scripts, Plans, Goals and Understanding" (1977).

The Restaurant Script

This is the canonical example:

```
SCRIPT: Restaurant
-----
TRACK: Fancy-Restaurant (specific variation)

PROPS:   Tables, chairs, menus, food, money, waiter, kitchen

ROLES:   Customer (C), Waiter (W), Chef (Ch), Cashier

ENTRY CONDITIONS:
- Customer is hungry
- Customer has money

SCENE 1: Entering
  C PTRANS into restaurant
  C MBUILD where-to-sit
  C PTRANS to table
  C MOVE body to sitting-position
```

```
SCENE 2: Ordering
  W PTRANS to table
  W MTRANS menu to C
  C MBUILD what-to-order
  C MTRANS order to W
  W PTRANS to kitchen
  W MTRANS order to Ch

SCENE 3: Eating
  Ch INGEST (prepares food)
  W PTRANS food to C
  C INGEST food

SCENE 4: Leaving
  W MTRANS bill to C
  C ATRANS money to W or cashier
  C PTRANS out of restaurant

EXIT CONDITIONS:
  - Customer is not hungry
  - Customer has less money
  - Restaurant has more money
```

Why Scripts Are Powerful

When an AI reads: *"John went to a restaurant. He left a big tip."*

Without a script, the AI can't connect these sentences.

With the restaurant script, the AI can infer:

- John was hungry (entry condition)
- John ate food (Scene 3)
- The service was apparently good (explanation for big tip)
- John paid (Scene 4)

Scripts allow AI to fill in the blanks — to "understand" by recognizing familiar patterns.

Script Variations

Scripts have:

- **Tracks:** Variations (fancy restaurant vs. fast food)
- **Default values:** What usually happens if not specified

- **Open slots:** What must be filled in (what he ordered)
- **Interference conditions:** What might disrupt the script (restaurant out of food)



Math Intuition Corner

Set Theory Foundation of Semantic Nets

Semantic network IS-A and AKO hierarchies correspond to set theory:

If Eagle IS-A Bird and Bird IS-A Animal:

$$\text{Eagle} \subseteq \text{Bird} \subseteq \text{Animal}$$

Any property P of Animal:

$$\forall x \in \text{Animal}: P(x) \text{ is true}$$

$$\text{Since } \text{Eagle} \subseteq \text{Animal}: \forall x \in \text{Eagle}: P(x) \text{ is true}$$

Inheritance = subset transitivity!



Video Resources

Topic	Video	Why Watch
Semantic Networks	Semantic Networks — AI lecture	Visual walkthrough
Frames in AI	Frames and Schemas	Clear explanation
Knowledge Representation overview	Knowledge Rep — MIT OpenCourseWare	Academic depth
Conceptual Dependency	Roger Schank on Scripts and CD	From the creator himself!



Worked Examples

Example 1: Build a Semantic Network

Domain: Computer Science classes

Knowledge:

- Data Structures IS-A Course
- Algorithms IS-A Course
- Algorithms AKO Data-Structures (algorithms builds on it)
- Algorithms HAS-TOPIC: Sorting, Searching, Graphs
- Data Structures HAS-TOPIC: Arrays, Trees, Linked-Lists
- Course IS-A University-Subject
- University-Subject HAS: Credits, Grade

Draw: Connect these as a semantic net with labeled arcs.

Inference from the net:

- What topics does Algorithms have? → Sorting, Searching, Graphs (direct)
- Does Algorithms have Credits? → Yes (inherited: Algorithms IS-A Course IS-A University-Subject HAS Credits)

Example 2: Create a Frame

Create a frame for a University Student:

```
FRAME: University-Student


---


IS-A:      Person
Name:      (required, string)
Student-ID: (required, unique integer)
Semester:  (value: 1-8, default: 1)
Courses:   (multiple values allowed)
CGPA:      (float 0.0-4.0, computed from grades)
Status:    (options: Active/Inactive/Graduated)

Slot: CGPA
  IF-NEEDED: Calculate from grade history

Slot: Status
  IF-ADDED: If Status = Graduated, archive record


---


```

Example 3: Apply a Script

Story: "Maria went to the bank. She waited in line. Then she handed the teller her check and got cash."

Script: Withdraw-Money-from-Bank

Without the script, each sentence is independent.

With the script, the AI fills in:

- Maria had an account
- Maria filled out a withdrawal form (implicit)
- The teller verified her identity (implicit)
- Maria now has less money in her account (exit condition)

⚡ Exam-Ready Summary

4 Approaches to Knowledge Representation

Approach	Method	Best For
Logical	FOPL sentences	Formal reasoning, proofs
Procedural	Algorithms, programs	"How to do" knowledge
Network	Semantic nets	Taxonomies, relationships
Structured	Frames, Scripts	Stereotyped situations

Semantic Networks: Key Points

- Nodes = concepts, Edges = labeled relationships
- IS-A enables inheritance (properties flow downward)
- Most-specific rule wins (exceptions override inherited properties)
- AKO = subclass relationship between classes

Frames: Key Points

- Template for a stereotyped concept or situation
- Has **slots** (attributes) with **facets** (metadata)

- Supports inheritance through IS-A hierarchy
- **Demons** (IF-ADDED, IF-NEEDED) add active behavior

Scripts: Key Points

- Represent sequences of events in typical situations
- Components: Props, Roles, Entry/Exit conditions, Scenes
- Enable understanding through pattern recognition and default filling
- Developed by Roger Schank

Conceptual Dependencies: Key Points

- ALL meanings reducible to ~12 primitive actions
 - Enables language-independent reasoning
 - Key primitives: ATRANS, PTRANS, MTRANS, INGEST, PROPEL
-

Chapter 5 Checklist

- ☐ Explain what a semantic network is and draw one with IS-A and HAS relationships
 - ☐ Demonstrate inheritance in a semantic network
 - ☐ Explain how exceptions are handled in semantic nets
 - ☐ Create a frame for a given concept with appropriate slots and facets
 - ☐ Explain the role of demons (IF-ADDED, IF-NEEDED)
 - ☐ Write out the restaurant script from memory
 - ☐ Name 5 CD primitives and give examples
 - ☐ Explain the frame problem and why it's important
 - ☐ Compare semantic nets vs. frames vs. scripts
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